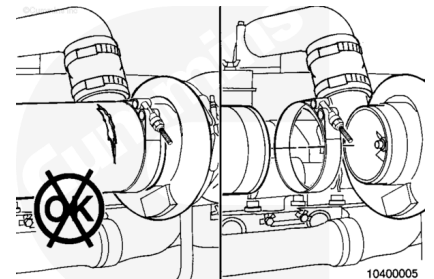


Trainings ▾

Inspect for Reuse

⚠ CAUTION ⚠

The engine air intake must be filtered to prevent dirt and debris from entering the engine. If intake air piping is damaged or loose, unfiltered air can enter the engine and cause premature wear.



Inspect the system for damaged or loose intake piping.

Replace damaged pipes.

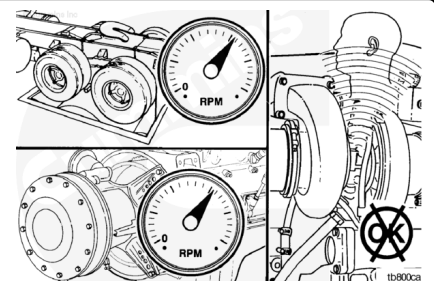
Tighten loose clamps.

Torque Value: 8 n•m [71 in-lb]

Stall speed is **not** full power.

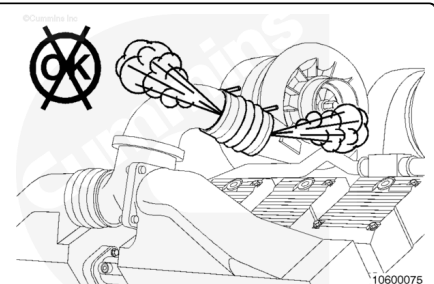
Operate the engine at rated rpm and load.

Listen for a high-pitched noise from the turbocharger.



This type of noise can be caused by an air leak from the following:

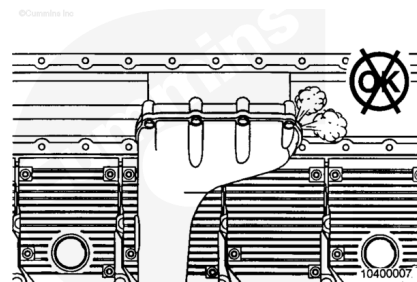
- Turbocharger-to-aftercooler air supply hose leak
- Air crossover-to-aftercooler connection leak
- Air crossover-to-intercooler connection leak
- Turbocharger-to-exhaust manifold gasket leak
- Turbine housing-to-bearing housing sealing surface leak
- Compressor housing-to-bearing housing sealing surface leak.



If a turbocharger-to-aftercooler air supply hose leak is found, inspect the hose for damage.

Replace damaged hose. Refer to Procedure 010-008
(</qs3/pubsys2/xml/en/procedures/56/56-010-008.html>).

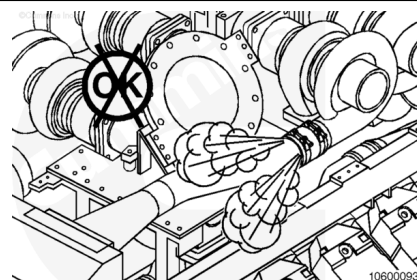
If a air crossover-to-aftercooler connection leak is found, replace the air crossover gasket.



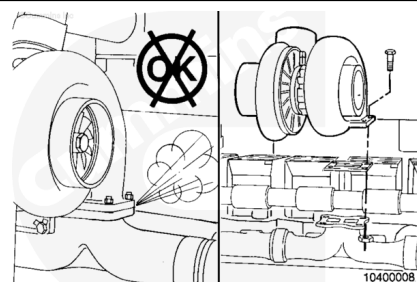
If a air crossover-to-intercooler connection leak is found, inspect for damaged hoses or loose hose clamps.

Replace damaged hoses.

Tighten loose clamps.



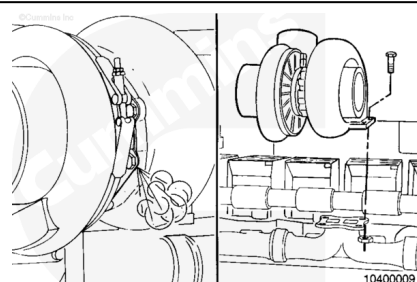
If a turbocharger-to-exhaust manifold gasket leak is found, replace the turbocharger base gasket.



If a turbine housing-to-bearing housing sealing surface leak is found, tighten any loose capscrews.

Torque Value: 20 n•m [15 ft-lb]

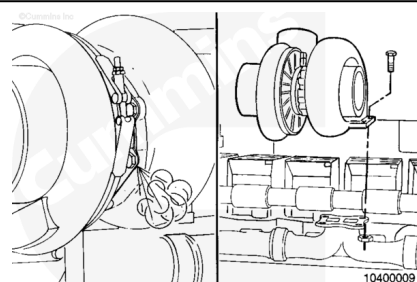
If an air leak is still present, replace the turbocharger.



If a compressor housing-to-bearing housing sealing surface leak is found, tighten the V-band clamps.

Torque Value: 8 n•m [71 in-lb]

If an air leak is still present, replace the turbocharger.



Pressure Test

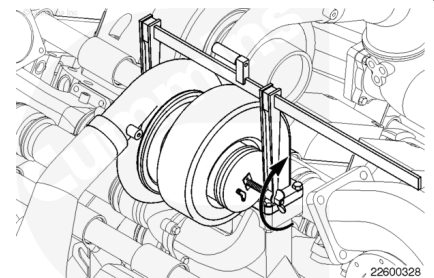
On-Engine Repair

- Remove the turbocharger heat shield if required, refer to Procedure 010-076 (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>)
- Remove the exhaust manifold heat shield if required, refer to Procedure 011-015 (</qs3/pubsys2/xml/en/procedures/56/56-011-015-tr.html>)
- Remove the air inlet piping from the turbocharger, refer to Procedure 010-130 (</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>)
- Remove the exhaust outlet piping from the turbocharger, refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html>).



⚠ CAUTION ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



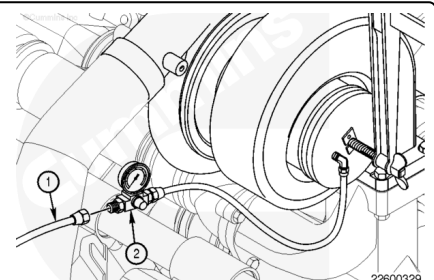
Check the condition of the seals prior to installing the exhaust manifold leak tester, Part Number 4918234. Replace seals if necessary.

With the turbocharger on the engine, install the clamp with the two seals centered over both sides of the turbocharger's air inlet and exhaust outlet connections.

Tighten the wing nut until the clamp seals are slightly compressed on the turbocharger inlet and outlet.

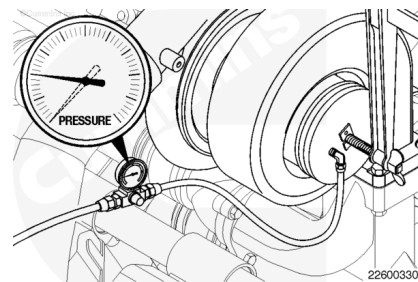
Connect the shop air hose (1) to the inlet air pressure regulator (2). Verify the air pressure regulator, Part Number 3164231, is at 0 kPa [0 psi] before continuing.

Connect the hose to the fitting on the turbocharger's exhaust outlet side of the exhaust manifold tester.

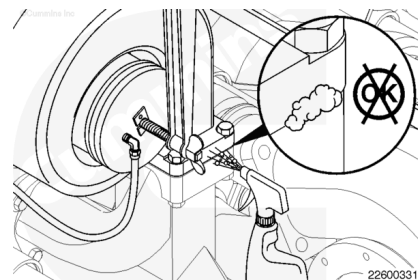


Note : The pressure will decrease slowly due to open valves in the combustion chamber.

Slowly adjust the air pressure regulator until the gauge indicates 69 kPa [10 psi] maximum pressure.

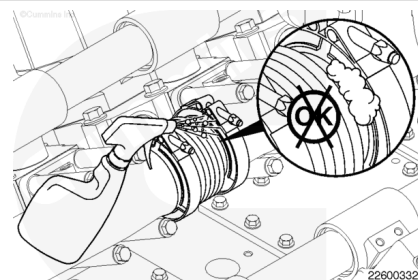


Spray a soap and water mixture onto the turbocharger's exhaust manifold mounting area. Check for bubbles, if bubbles are present, replace the turbocharger-mounting gasket. Refer to Procedure 010-034 (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>).



Note : Small amounts of bubbles are acceptable. These will seal during the engine operation as the bellows expand at operating temperature.

Spray a soap and water mixture onto the exhaust bellows. Check for bubbles, if large bubbles exist, tighten the exhaust bellow clamp(s) and repeat the pressure test procedures.

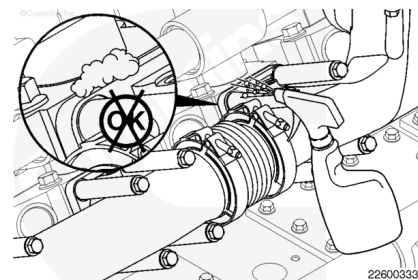


Torque Value: 17 n•m [150 in-lb]

If large amounts of bubbles still exist, replace the exhaust bellow gasket(s). Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>) and repeat the pressure test procedures.

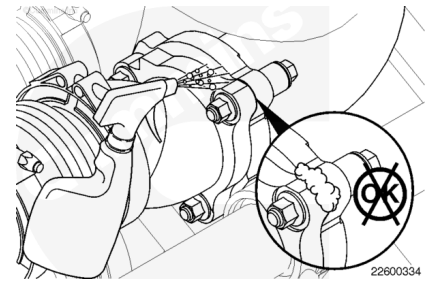
For engines with exhaust bellows, spray a soap and water mixture onto the cylinder head next to the exhaust manifold gasket mounting area and surrounding exhaust system components.

Check for bubbles, if bubbles exist, replace the gasket(s). Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>) and repeat the test procedures.



For engines with exhaust bellows, spray a soap and water mixture on the exhaust manifold adapter gasket and mounting area.

If large amounts bubbles exist, replace the exhaust manifold adapter gasket. Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>) and repeat the test procedures.

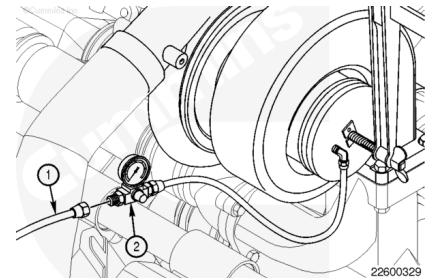


Make certain that the air pressure regulator is set at 0 kPa [0 psi] before removing the clamp, air pressure regulator, and hose.

Remove the shop hose (1) from the inlet air pressure regulator.

Remove the hose from the fitting on the exhaust manifold tester (2).

Remove the clamp from the turbocharger.



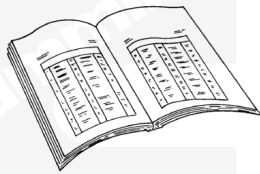
- Install the exhaust outlet piping to the turbocharger, refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html>)
- Install the air inlet piping to the turbocharger, refer to Procedure 010-130 (</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>)
- Install the exhaust manifold heat shield if required, refer to Procedure 011-015 (</qs3/pubsys2/xml/en/procedures/56/56-011-015-tr.html>)
- Install the turbocharger heat shield if required, refer to Procedure 010-076 (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>).



Pressure Test

Off-Engine Repair

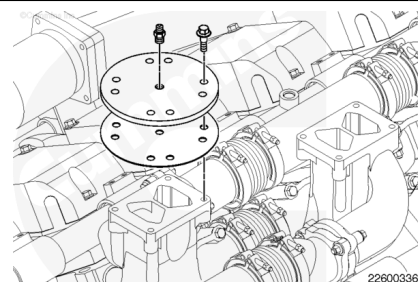
- Remove the turbocharger, refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html>).



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With the turbocharger removed, locate the correct mounting pattern on the pressure test plate (1) and the rubber seal (2) from the exhaust manifold leak test kit, Part Number 4918187.

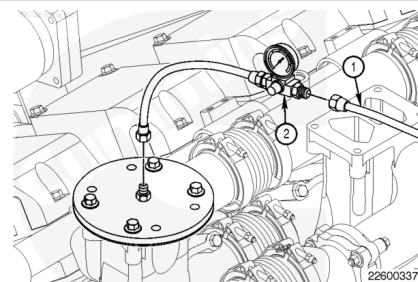
Install the rubber seal and the pressure test tool plate onto the exhaust manifold mounting flange. Install the appropriate hardware to hold down the pressure test plate and rubber seal to exhaust manifold. Tighten the capscrews.



Torque Value: 20 n·m [15 ft·lb]

⚠ CAUTION ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Connect the shop air hose (1) to the inlet air pressure regulator (2). Verify the air pressure regulator, Part Number 3164231, is at 0 kPa [0 psi] before continuing.

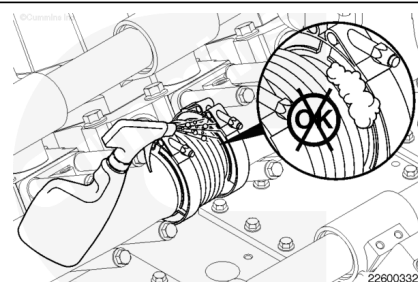
Connect the hose to the fitting on the turbocharger's exhaust outlet side of the exhaust manifold tester.

Note : The pressure will decrease slowly due to open valves in the combustion chamber.

Slowly adjust the air pressure regulator until the gauge indicates 69 kPa [10 psi] maximum pressure.

Note : Small amounts of bubbles are acceptable. These will seal during the engine operation as the bellows expand at operating temperature.

Spray a soap and water mixture onto the exhaust bellows. Check for bubbles. If large bubbles exist, tighten the exhaust bellow clamp(s) and repeat the pressure test procedures.

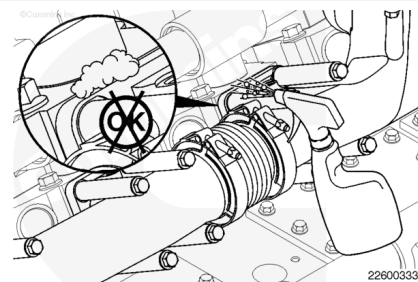


Torque Value: 17 n·m [150 in·lb]

If large amounts of bubbles still exist, replace the exhaust bellow gasket(s). Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html>) and repeat the pressure test procedures.

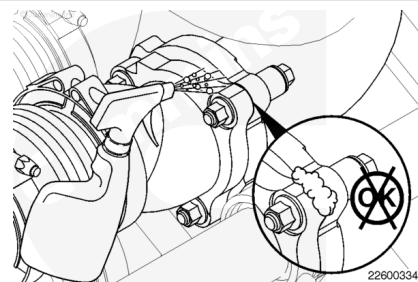
For engines with exhaust bellows, spray a soap and water mixture onto the cylinder head next to the exhaust manifold gasket mounting area and surrounding exhaust system components.

Check for bubbles. If bubbles exist, replace the gasket(s), refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>) and repeat the test procedures.



On engines with exhaust bellows and exhaust manifold adapter gaskets, spray a soap and water mixture surrounding the mounting area.

If large amounts bubbles exist, replace the exhaust manifold adapter gasket. Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>) and repeat the test procedures.

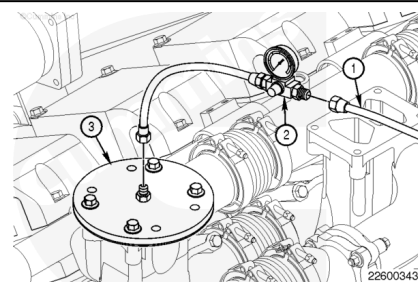


Make certain that the air pressure regulator is set at 0 kPa [0 psi] before removing the clamp, air pressure regulator, and hoses.

Remove the shop hose (1) from the inlet air pressure regulator (2).

Remove the hose from the fitting on the pressure test tool plate (3).

Remove the rubber seal, pressure test plate, capscrews, and nuts from exhaust manifold mounting flange.



- Install the turbocharger, refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/56/56-010-033-tr.html>)



